

AYMANE AIT DADS

Machine Learning Engineer | Satellite-Derived Bathymetry · Multispectral Fusion · Geospatial ML Research
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PROFESSIONAL SUMMARY

Master-level engineering candidate at EURECOM with demonstrated expertise in training ML regression and classification models on large-scale multispectral and structured datasets, directly applicable to fusing Sentinel-2 imagery with ICESat-2 depth points for Satellite-Derived Bathymetry. Ranked top 10% on a \$106K+ Kaggle competition leaderboard (0.80/1.0) fine-tuning a 30B Mamba-Transformer MoE model with LoRA, demonstrating rigorous ablation study methodology and scalable ML engineering. Core competencies span Random Forest, Neural Networks, ensemble methods, feature engineering, and unsupervised clustering - all cited in the SDB assignment literature. Available immediately for MSc thesis internship in Europe; no visa sponsorship required.

CORE COMPETENCIES

Machine Learning Fusion | Multispectral Imagery | Random Forest Regression | Ablation Studies | Scalable ML Workflow | Optical Depth Retrieval | Neural Networks | Unsupervised Clustering

WORK EXPERIENCE

Orange Maroc Jul 2024 - Aug 2024
Data Science Intern - Casablanca, Morocco

- Built **Random Forest and K-Means clustering models** on 100,000+ fiber-optic network records to classify signal quality across thousands of geographic nodes, directly mirroring ML regression workflows for spatial data.
- Developed an end-to-end clustering pipeline that mapped customer performance profiles to **5 distinct service tiers**, producing outputs immediately adopted by the network optimization engineering team.
- Reduced manual analysis time by **~60%** through automated feature engineering and model evaluation pipelines built with Scikit-learn and Pandas.
- Delivered stakeholder presentations translating model outputs and spatial quality maps into actionable maintenance recommendations for **non-technical engineering audiences**.

PROJECTS

NVIDIA Nemotron Reasoning Challenge - Kaggle Kaggle · In Progress Jun 2026 · Top 10% Public Leaderboard · \$106K+ Prize Pool
Competition

- Achieved **top 10% public leaderboard score (0.80/1.0)** in a \$106K+ prize-pool competition by fine-tuning a 30B Mamba-Transformer MoE model with LoRA (r=32, ~888M trainable parameters) on 7,828 verified chain-of-thought reasoning examples using SFTTrainer with completion-only loss.
- Resolved **5 Blackwell GPU (sm_120) incompatibilities** in strict dependency order - including monkey-patching CUDA allocators, stubbing Mamba/Cutlass imports, and replacing Triton kernels with PyTorch fallbacks - and conducted 20+ single-variable ablations confirming that completion-only loss (0.80) outperforms full-sequence loss (0.79) and that sequence packing degrades generalization (0.80 to 0.64).

PyTorch · Unsloth · PEFT · Hugging Face Transformers · vLLM · Triton

Aerial Cactus Detection - Endangered Species Classification EURECOM · 2025 · 99.8% Accuracy · ROC AUC 1.0

- Benchmarked CNN, DenseNet121, and hybrid CNN-Transformer architectures on **17,500+ drone images** for endangered species classification, achieving 99.8% accuracy, F1 = 0.999, and ROC AUC = 1.0 - methodology directly transferable to multispectral satellite image classification.
- Designed a rigorous comparative evaluation framework across three model families, producing **scalable inference workflows** applicable to geospatial remote sensing tasks such as Sentinel-2 optical depth retrieval.

PyTorch · DenseNet121 · Scikit-learn · Pandas · Matplotlib

AI-Generated Image Detection - NTIRE 2026 @ CVPR EURECOM ImSecu + NTIRE 2026 @ CVPR · 2025 · Ranked 1st EURECOM Class Leaderboard (0.791 AUC)

- Ranked **1st on the EURECOM private class Kaggle leaderboard (0.791 AUC)** by fine-tuning CLIP ViT-L/14 (428M parameters) with a custom MLP head and dual learning rates on 250,000 images spanning 25+ generator types; also submitted independently to the NTIRE 2026 @ CVPR CodaBench international benchmark.
- Validated that **1-epoch training (0.793 AUC) outperforms 4-epoch training (0.767 AUC)** through systematic ablation of freeze depth and training duration, then built a 3-model ensemble with 10-view TTA to maximize out-of-distribution generalization - directly applicable to environmental variability (turbidity, bottom type) in SDB accuracy assessment.

PyTorch · OpenCLIP · ViT · FP16 Mixed Precision · Kaggle GPU

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Statistics, Computer Vision, NLP, Image Security, Cloud Computing, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

ML & Geospatial Modeling: Random Forest | Neural Networks | Scikit-learn | Ensemble Methods | Feature Engineering | Clustering | Ablation Studies

Deep Learning & Computer Vision: PyTorch | CLIP / ViT | DenseNet | CNN | Hugging Face Transformers | FP16 Mixed Precision | TensorFlow

Data Engineering & Infrastructure: Python (expert) | Pandas | NumPy | SQL | Power BI | Git | Linux | Docker (basic)